

EQUITY RESEARCH | SEMICONDUCTORS | PHOTOMASKS

Photronics, Inc.

NASDAQ: PLAB

The Photomask Shovel-Seller: AI Infrastructure's Most Overlooked Enabler

Initiation of Coverage | June 2026 | 12-Month Price Target: \$46.90

RATING	PRICE TARGET	CURRENT PRICE	UPSIDE
BUY	\$46.90	~\$30,73	~52,6%

KEY METRICS	VALUE
Ticker / Exchange	PLAB / NASDAQ
Market Cap (est.)	~\$1.88B
Revenue TTM FY2025	\$849M
Revenue Q2 FY2026 (quarterly)	\$209.9M
Gross Margin Q2 FY2026	31.3% (vs 36.9% Q2-25)
Operating Margin Q2 FY2026	20.1%
Non-GAAP EPS Q2 FY2026	\$0.42
Net Cash (est. PLAB own)	~\$160M (excl. JV cash)
Total Cash + ST Investments	\$637.7M
Diluted Shares	58.7M
FY2026 CapEx Guidance	\$330M (reaffirmed)
Fiscal Year End	Late October
Headquarters	Brookfield, CT
CEO	George Macricostas

Investment Summary: Photronics is a listed pure-play photomask manufacturer — provider of the quartz "stencils" used to print every semiconductor circuit and flat-panel display. The company's fortress balance sheet (\$637M cash, negligible debt), disciplined buyback programme (shares down ~3.5% YoY), and strategically timed \$330M capex cycle position it as an under-covered shovel seller of the AI infrastructure build-out. We initiate with BUY and a 12-month DCF-derived price target of \$46.90, implying ~52,6% upside.

IMPORTANT: This report is for informational purposes only and does not constitute investment advice. All estimates are our own. See full disclosures on final page. Readers should verify all primary sources before making investment decisions.

What Photronics Does	3
Management.....	3
2. Technology & Products	3
IC Photomasks — Core Business (~70% Revenue).....	3
FPD Photomasks — Growth Driver (~30% Revenue).....	4
Product Revenue Mix	4
3. Business Model	4
Revenue Mechanics.....	4
Cost Structure	4
4. Competitive Landscape.....	5
5. Industry Context: Semiconductor Supply Chain 2026.....	5
Three Simultaneous Bottlenecks.....	5
6. CapEx Programme — \$330M FY2026	6
Official CapEx Categories (per Investor Presentation, May 2026).....	6
Key Earnings Call Quotes on CapEx	6
7. Financial Summary.....	7
Income Statement — Recent Quarters	7
Key Balance Sheet Items (3 May 2026)	7
Management Guidance — Q3 FY2026	7
8. Valuation — DCF & Scenario Analysis	8
DCF Framework.....	8
DCF Summary — Base Case	8
Revenue Scenarios (USD Millions).....	8
9. Risks & Catalysts.....	9
Key Risks	9
Key Catalysts.....	9
10. Investment Thesis — BUY	9
Metrics to Monitor Quarterly	10
11. Sources & Disclosures.....	10
Primary Sources — SEC EDGAR (Verified)	10
Secondary Sources (Analyst / Industry)	10
Important Disclosures	11

1. COMPANY OVERVIEW

What Photronics Does

Photronics, Inc. (NASDAQ: PLAB), founded in 1969, is the world's largest independent (merchant) manufacturer of photomasks — ultra-precise quartz templates that transfer circuit patterns onto silicon wafers during semiconductor fabrication. Every chip produced today, from the most advanced TSMC 3nm GPU to a commodity 180nm automotive microcontroller, requires between 30 and 90 unique photomasks (one per lithographic layer). Without photomasks, there are no chips.

Photronics operates 11 manufacturing plants across the US, Europe, Taiwan, South Korea and China, making it the only merchant supplier with true global geographic coverage. It is the sole listed pure-play in the sector: primary competitors Toppan Photomasks and DNP (Dai Nippon Printing) are buried within large Japanese industrial conglomerates, offering investors no direct exposure.

In our view, geographic diversification is one of Photronics' strongest competitive advantages. As the only listed pure-play photomask provider with manufacturing presence across the main semiconductor regions — Asia, Europe and North America — the company is uniquely positioned to serve customers seeking resilient and geographically diversified mask supply.

JV Structure — Key Complexity

Two critical joint ventures underpin the Asia business:

- PDMC (Photronics-DNP Mask Corp, Taiwan) — 50.01% PLAB / 49.99% DNP.
- PSMC (Photronics Semiconductor Mask Corp, South Korea) — 50.01% PLAB / 49.99% DNP.

Photronics' regional JV structure strengthens its access to key Asian semiconductor ecosystems, particularly in Korea and Taiwan, where proximity to major customers such as Samsung and TSMC is strategically valuable. While these companies operate captive mask-making capabilities, periods of capacity tightness or overflow demand can create opportunities for nearby qualified merchant suppliers such as Photronics.

Photronics consolidates 100% of JV revenue/expenses (control basis) but retains only 50.01% of net income. Of the \$637.7M in cash on the Q2 FY2026 balance sheet, approximately \$477M resides within the JVs and is not freely distributable to PLAB shareholders. We estimate PLAB-own cash at ~\$160M. Which Will be usefull for DCF.

Management

George Macricostas	Chairman & CEO — son of founder, multi-decade tenure
Christopher J. Lutzo	VP, General Counsel & Secretary
Ted Moreau	VP, Investor Relations

2. TECHNOLOGY & PRODUCTS

IC Photomasks — Core Business (~70% Revenue)

IC photomasks are used in semiconductor lithography: a light source shines through the mask, projecting the circuit pattern onto a photo-sensitive wafer. Each design layer requires a distinct mask. Photronics focuses on mainstream nodes (28nm and above) where it holds a dominant market position, serving foundries such as UMC, GlobalFoundries, SMIC, X-FAB and Tower, plus IDMs like Texas Instruments, Analog Devices, Microchip and ON Semiconductor.

Entry barriers are formidable: a single e-beam mask-writing tool costs \$20-50M; a full-scale mask plant costs \$200M+. Qualification cycles with foundries take 12-18 months. Photronics holds ISO/TS 16949 (automotive) and ISO 13485 (medical) certifications, enabling it to serve safety-critical applications.

FPD Photomasks — Growth Driver (~30% Revenue)

Flat-panel display (FPD) photomasks are used to manufacture LCD and OLED screens. This segment grew +13% YoY in Q2 FY2026 while IC declined -5%. Drivers: global shift from LCD to OLED (more complex masks), larger generation glass (Gen 10.5+) for TV-sized panels, and AMOLED G8.6 expansion. Photronics holds a leading position in AMOLED masks acknowledged by management in earnings calls.

Product Revenue Mix

Segment	FY2025 Rev (est.)	% Mix	Gross Margin (est.)	YoY Q2-26
IC High-End (≤28nm)	\$227M	27%	38-45%	—
IC Mainstream (>28nm)	\$364M	43%	27-34%	-5%
FPD High-End (G10.5+/AMOLED)	\$211M	25%	34-42%	+13%
FPD Mainstream (LCD)	\$38M	5%	15-25%	—
TOTAL	~\$840M	100%	~33%	-0.5% YoY

Source: Company filings, analyst estimates. FY2025 segment split estimated from TTM proxy. Gross margins are estimates — PLAB does not publish segmented margin disclosure.

3. BUSINESS MODEL

Revenue Mechanics

Photronics operates a make-to-order model: a customer submits a chip design; Photronics fabricates the corresponding mask set within 1-7 days depending on complexity; revenue is recognised at delivery. No royalties or wafer-consumption fees.

Repeat demand: once a chip design enters volume production, the customer periodically reorders identical masks as backups or to support multiple fabs, creating recurring revenue independent of new tape-outs. With ~675 active customers — the top three representing ~37% of revenue — the business is well-diversified.

Cost Structure

Cost Line	% Revenue Q2-26	Comment
Cost of Goods Sold	69%	Materials, labour, utilities, D&A
SG&A	10%	Global sales force + administration
R&D (Q2 Results)	1.3%	Low vs tech peers — sector maturity; declining recently
CapEx FY2026 (guidance)	~39%	\$330M peak; normalises to \$160-180M from FY2028+

4. COMPETITIVE LANDSCAPE

Company	Type	Advanced/EUV	Mainstream	FPD	Global Reach	EV/EBITDA
Photronics (PLAB)	Merchant (Listed)	No (8nm push)	Focus	Strong	US/EU/TW/KR/CN	4,4x
Tekscend	Pure play (JPN co.)	Yes (dominant)	Strong	Limited	Asia-focused	5,7x
DNP Photomasks	Merchant (JPN co.)	Yes (dominant)	Strong	Limited	Asia-focused	8,2x
TSMC Captive	Captive	Yes	Yes	N/A	Taiwan	14,7x
Samsung Captive	Captive	Yes	Yes	N/A	Korea+	7,7x
Intel Captive	Captive	Yes	Yes	N/A	US/EU	15,7x

We believe Tekscend is the closest operational comparable to Photronics, as it is one of the few pure-play merchant photomask providers. Other industry participants, such as captive mask shops within leading foundries and IDMs, are exposed to photomask production but primarily serve their own internal demand rather than the merchant market.

As discussed earlier, Photronics' global manufacturing footprint represents a meaningful competitive advantage. Its presence across the major semiconductor regions allows customers to outsource mask production locally, reduce logistics risk, avoid potential delays from trade restrictions or tariffs, and build greater supply-chain redundancy.

5. INDUSTRY CONTEXT: SEMICONDUCTOR SUPPLY CHAIN 2026

Three Simultaneous Bottlenecks

Bottleneck	Status Q2-26	Impact on PLAB	Resolution estimation
HBM/DRAM/NAND Memory	SK Hynix & Samsung sold-out 2026-27. DRAM prices +300% YoY est. (Goldman Sachs)	NEGATIVE near-term: OEM delays reduce tape-outs. IC Mainstream -6% YoY Q2-26.	2027-2028
High utilization rates (2-3nm)	TSMC 2nm 100% booked 2026-27. Samsung SF2 yield ~40% vs TSMC 65%.	NEUTRAL: PLAB not in EUV. AI infrastructure demand indirectly supportive.	2028+
Advanced Packaging (CoWoS)	TSMC lead times 50-78 weeks. NVIDIA captures 50-70% of CoWoS capacity.	POSITIVE: Packaging masks use mainstream nodes — PLAB sweet spot.	2026-27 partial

CEO quote (Q2 FY2026 earnings, 28-may-2026): "certain design releases have been delayed due to elevated fab utilization rates, which are extending new product launch timelines, memory supply constraints and related cost pressures for OEMs, and geopolitical uncertainty."

Memory manufacturers are prioritizing high-margin AI-related products such as HBM, which is tightening supply for conventional DRAM and other non-HBM memory products. This creates cost pressure and potential supply constraints for OEMs, delaying new product launches and, ultimately, some design releases.

Elevated foundry utilization rates are delaying the release of new designs, as fabs prioritize existing high-volume production over new product introductions. This constrains near-term photomask demand for Photronics, since revenue is primarily driven by design releases and mask orders rather than wafer volumes.

6. CAPEX PROGRAMME — \$330M FY2026

Photronics has reaffirmed \$330M in CapEx for FY2026 — 39% of FY2025 revenue and nearly 2× historical run-rate. Per the May 2026 Investor Presentation (slide 11), PLAB describes three official capex categories: (1) Allen TX, (2) Korea 8nm, (3) End-of-life tool upgrades.

Official CapEx Categories (per Investor Presentation, May 2026)

Category (official)	Status (Q2 FY2026)	First Revenue	Sustainable Run-Rate
Allen, TX — US plant expansion	Latter stages of expansion. Qual. masks Q3 FY2026.	Q4 FY2026 (small)	FY2027+ material
Korea — 8nm cleanroom extension	Initial expansion. Equipment install H2 FY2026.	FY2027	FY2028+ material
End-of-life tool upgrades	Peak in FY2026 (official quote).	N/A (maintenance)	~\$65M est. sustainable

Note: PLAB additionally mentions "Multi-Beam Mask Writer expands technical and production capability through improved cycle time and extends complex patterning opportunities" (Aug 2025 Investor Presentation, slide 14) — this refers to the Boise, Idaho installation announced 8-Aug-2025. PLAB does not assign it a separate capex bucket; it is part of the US expansion envelope.

Key Earnings Call Quotes on CapEx

"At the Allen facility, Photronics is beginning production of qualification masks and continues to target initial revenue late in fiscal 2026, with a more meaningful revenue contribution expected in 2027 and beyond." — Q2 FY2026 earnings call, 28-may-2026

"In Korea, the company is preparing cleanroom space for equipment that will extend capabilities to 8nm and below, with installations expected to begin later in the fiscal year." — Q2 FY2026 earnings call, 28-may-2026

"The company continues to recognize growth for masks that support exciting areas such as AI-driven chip packaging applications and masks for high NA EUV development projects." — Q2 FY2026 earnings call, 28-may-2026

7. FINANCIAL SUMMARY

Income Statement — Recent Quarters

USD '000	Q2 FY2026	Q1 FY2026	Q2 FY2025	YoY Change
Revenue	\$209,940	\$225,066	\$210,992	-0.5%
Gross Profit	\$65,757	\$78,702	\$77,906	-15.6%
Gross Margin	31.3%	35.0%	36.9%	-560 bps
Operating Income	\$42,179	\$54,859	\$55,717	-24.3%
Operating Margin	20.1%	24.4%	26.4%	-630 bps
Net Income (consol.)	\$43,250	\$60,177	\$24,217	+78.6%*
NCI — to DNP	(\$11,821)	(\$17,238)	(\$15,356)	n/m
NI attrib. PLAB	\$31,429	\$42,939	\$8,861	+255%*
GAAP EPS (diluted)	\$0.54	\$0.74	\$0.15	+260%*
Non-GAAP EPS	\$0.42	\$0.61	\$0.40	+5.0%
Diluted Shares (M)	58.7	58.4	61.0	-3.8%

* YoY comparisons distorted by FX gains/losses. Non-GAAP EPS strips FX and is the most stable metric. Source: Exhibit 99.1, 8-K Q2 FY2026.

Key Balance Sheet Items (3 May 2026)

Cash + ST Investments	\$637.7M — of which \$477M in JVs (~\$160M freely available to PLAB)
PP&E (net)	\$927.9M
Long-Term Debt	\$3.9M — virtually debt-free
PLAB shareholders' equity	\$1,241M
Total Assets	\$1,931M

Management Guidance — Q3 FY2026

Revenue	\$207M – \$215M (broadly flat vs Q2)
Operating Margin	18% – 20% (modest compression continues)
Non-GAAP EPS	\$0.39 – \$0.45
Diluted Shares	~59M
FY2026 Full-Year CapEx	~\$330M (reaffirmed)

8. VALUATION — DCF & SCENARIO ANALYSIS

DCF Framework

10-year explicit FCF model segmented across four revenue streams (IC High-End, IC Mainstream, FPD High-End, FPD Mainstream) under three scenarios. FCF adjusted for NCI (~50% of JV-derived FCF flows to DNP). Dual-stage WACC (10.2% during peak CapEx, declining to 8.75% as business matures). Terminal growth rate: 3.0% (Gordon Growth Model).

DCF Summary — Base Case

WACC (FY2026 peak)	10.2%
WACC (long-term)	8.75%
Terminal Growth Rate	3.0%
PV of Explicit FCFs (10Y)	\$715,4M
PV of Terminal Value	\$1,640,8M
Enterprise Value	\$2,356,2M
+ PLAB-own Cash (est.)	\$399M
- Debt	\$4M
Equity Value	\$2,751,1M
Diluted Shares	58.7M
Fair Value per Share	\$46.87
Current Price (est.)	~\$32.00
Implied Upside	~46%

Revenue Scenarios (USD Millions)

Scenario	FY2026	FY2027	FY2028	FY2030	FY2035	CAGR '26-'35
Bear	\$868M	\$932M	\$1,018M	\$1,243M	\$1,744M	~7.6%
Base	\$876M	\$956M	\$1,062M	\$1,357M	\$2,027M	~9.6%
Bull	\$884M	\$981M	\$1,111M	\$1,488M	\$2,415M	~11.6%

9. RISKS & CATALYSTS

Key Risks

- Margin compression: Gross margin fell from 36.9% (Q2-25) to 31.3% (Q2-26) — 560bps contraction. If CapEx-driven cost increases persist beyond FY2027, terminal value assumptions weaken.
- No EUV exposure: PLAB does not compete in EUV mask market (~2nm/3nm), capping relative TAM versus Toppan and DNP.
- JV cash opacity: \$477M of \$637M cash is JV-trapped; reported cash overstates freely available liquidity by ~3x.
- Cyclicity: Photomask demand lags the semiconductor cycle by 2-4 quarters. Extended memory or foundry downturns directly compress revenue.
- Customer concentration: Top 3 customers ~37% of revenue. Loss of one is material.
- Geopolitical / tariff risk: ~75% of revenue is Asia-sourced. US-China escalation or Taiwan strait tensions are tail risks.
- FY2026 FCF negative: \$330M CapEx > ~\$220M OCF estimate. FCF negative in FY2026.
- Allen TX execution: If TSMC/Intel delay their US fabs, Allen demand timeline shifts right.
- Korea Samsung concentration: Samsung is the anchor for the 8nm JV — if Samsung reduces outsourcing, the investment is underutilised.

Key Catalysts

- AI packaging (near-term): Multi-beam writer in Boise operational since Aug-25. CoWoS/EMIB interposer mask demand accelerating — mainstream-node masks play directly to PLAB's strengths.
- Allen TX revenue ramp: First commercial masks Q4 FY2026; material contribution FY2027. Sole high-end capable merchant in US — geographic premium pricing upside.
- Korea 8nm (FY2027+): Moving up the value chain to advanced nodes multiplies ASP 3-5x vs mainstream. Samsung partnership via PSMC provides anchor demand.
- Memory shortage resolution (~2027-28): When HBM/DRAM supply normalises, delayed OEM tape-outs resume, restoring IC mainstream mask demand.
- CHIPS Act + EU Chips Act: Geographic diversification mandates drive local mask shop demand — PLAB has US plants ready.
- Buyback continuation: Shares down 3.5% YoY (60.8M → 58.7M). Continued repurchases are accretive.

10. INVESTMENT THESIS — BUY

RATING	PRICE TARGET	CURRENT PRICE	UPSIDE
BUY	\$46.90	~\$30.73	~52.6%

Photronics is the shovel seller of the semiconductor boom. It does not matter whether TSMC or Samsung wins the foundry war, whether NVIDIA or AMD leads in AI GPUs, or whether memory peaks in HBM3e or HBM4. Every new chip design — across every node, geography and application — requires Photronics's photomasks. That is the core of the investment case.

Why it is cheap: The market discounts PLAB for four reasons — (1) no EUV exposure limits the "AI narrative"; (2) near-term margin compression from the CapEx peak; (3) JV opacity overstates cash and complicates analysis (4) Memory bottlenecks as a headwind rather than an opportunity.

Each is a solvable transparency problem, not a structural impairment.

Why now: The CapEx cycle is past its peak commitment point. Allen Texas begins commercial revenue Q4 FY2026. AI packaging masks are already generating incremental revenue. Korea 8nm crystallises FY2027-28 as a genuine advance in competitive position. As these projects deliver, the market will reprice toward fair value. We see a 12-36 month path to \$46+ per share.

Metrics to Monitor Quarterly

- IC vs FPD revenue split — watch for IC recovery as memory normalises (~2027)
- Gross margin trajectory — target return to >35% by FY2027-28 is thesis-critical
- Operating margin vs guidance — recovery from 18-20% Q3-26 to 24-26% FY2028
- Allen Texas revenue contribution — first data point Q4 FY2026 earnings
- Korea 8nm equipment installation progress — confirming FY2027 revenue timeline
- AI packaging masks revenue — how much of the incremental revenue from Boise multi-beam
- Share count — continued buyback from 58.7M toward 56M+ is accretive signal
- JV cash dividends / PDMC-PSMC distributions — key to improving PLAB's own liquidity
- FX impact on EPS — NTD and KRW vs USD are material quarterly noise
- Tape-out activity signals from foundry customers — leading indicator of mask demand

11. SOURCES & DISCLOSURES

Primary Sources — SEC EDGAR (Verified)

All links below are direct SEC EDGAR filing URLs verified at time of writing:

[8-K Q2 FY2026 — Earnings release + Investor Presentation \(28-may-2026\)](#)
[Exhibit 99.1 — Press release Q2 FY2026 \(financials, guidance, CEO quote\)](#)
[Exhibit 99.2 — Investor Presentation mayo 2026 \(CapEx slides 11, 16, 17\)](#)
[8-K Q1 FY2026 — Earnings + Investor Presentation \(25-feb-2026\)](#)
[8-K Q3 FY2025 — Earnings + multi-beam writer press release \(27-aug-2025\)](#)
[8-K Q4 FY2025 \(10-dic-2025\)](#)
[10-K FY2025 — Annual Report \(17-dic-2025\)](#)
[10-K FY2024 — Annual Report \(19-dic-2024\)](#)
[SEC EDGAR — All PLAB filings](#)
[Photronics Investor Relations \(IR presentations, press releases\)](#)

Secondary Sources (Analyst / Industry)

The following sources are behind paywalls or are industry/forum estimates. Links point to homepages or closest public reference:

TrendForce — yields, memory pricing: [trendforce.com](https://www.trendforce.com)
 Digitimes — Samsung NAND→DRAM, YMTC: [digitimes.com](https://www.digitimes.com)
 Tom's Hardware — Intel 18A yields, HBM: [tomshardware.com](https://www.tomshardware.com)
 SemiWiki — Samsung Foundry utilization: [semiwiki.com](https://www.semiwiki.com) *[forum — not primary source]*
 Goldman Sachs — DRAM undersupply forecasts (via Benzinga): [benzinga.com](https://www.benzinga.com) *[paywall]*
 Omdia — FPD photomask market: [ondia.tech.informa.com](https://www.ondia.tech.informa.com) *[paywall]*

IMPORTANT DISCLOSURES

This equity research report was prepared for informational purposes only. It does not constitute an offer to buy or sell any security referenced herein. The information contained in this report has been compiled from sources believed to be reliable, including Photronics SEC filings (10-K FY2025, 8-K Q2 FY2026, investor presentations), earnings call transcripts, and industry research, but no representation or warranty is made as to its accuracy or completeness.

All price targets, earnings estimates and financial projections are forward-looking statements and involve known and unknown risks, uncertainties and other factors that could cause actual results to differ materially. Past performance is not indicative of future results. Investors should conduct their own due diligence and consult a qualified financial adviser before making investment decisions.

Data as of June 2026. Photronics fiscal year ends in late October.